

Charmgil Hong

558 Handong-ro, Buk-gu, Pohang, Gyeongbuk 37554 • +82-54-260-1309
charmgil@handong.edu • <https://charmgil.github.io/>

INTERESTS

- Developing and applying advanced machine learning techniques to solve complex, large-scale, and multimodal problems in medical and industrial domains
- Enhancing the quality and robustness of predictive models by integrating multiple data modalities and perspectives
- Building and applying state-of-the-art machine learning models for effective anomaly detection across diverse domains

EDUCATION

- Ph.D. in Computer Science** 2017
- Dissertation: *Multivariate Data Modeling and Its Applications to Conditional Outlier Detection*
Advisor: Dr. Milos Hauskrecht
University of Pittsburgh, Pittsburgh, PA, USA
- B.S.E. in Computer Science and Electronic Engineering** (with Distinction) 2010
Handong Global University, Pohang, Gyeongbuk, Korea

WORK EXPERIENCES

- Associate Professor** Mar 2025 - Present
School of Computer Science and Electrical Engineering, Handong Global University, Pohang, Gyeongbuk, Korea
- Teach courses in machine learning, databases, Linux environments, and computer programming.
- Design and implement curricular and extracurricular programs aimed at maximizing student learning outcomes in a teaching-focused institution; This includes developing educational programs, organizing and supervising competitions, and coordinating student internship/exchange opportunities.
- Lead the Handong Artificial Intelligence Lab (HAIL) research group, focusing on medical and industrial AI, forensic/surveillance technology, search technologies, and demand and price forecasting.
- Assistant Professor** Feb 2019 - Feb 2025
School of Computer Science and Electrical Engineering, Handong Global University, Pohang, Gyeongbuk, Korea
- Postdoctoral Associate** Oct 2017 - Dec 2018
Department of Computer Science, University of Pittsburgh, Pittsburgh, PA, USA
- Conducted predictive data modeling and outlier detection to prevent medical/clinical errors
- Analyzed electronic health records (EHRs) and develop machine learning methodologies
Supervisor: Dr. Milos Hauskrecht
- Graduate Student Researcher** May 2011 - Aug 2017
Department of Computer Science, University of Pittsburgh, Pittsburgh, PA, USA
- Primary focus: Multi-Label Classification that aims to learn and predict multi-dimensional output; Conditional Outlier Detection for clinical applications that monitor and alert on medical errors
- Analyzed and processed the EHRs that University of Pittsburgh Medical Center (UPMC) maintains; Developed software packages that help data preprocessing, feature generation for predictive modeling, outlier detection and alerting
Supervisor: Dr. Milos Hauskrecht

Research Intern

May 2015 - Aug 2015

Siemens Corporate Technology, Princeton, NJ, USA

- Topic: Sensory data analysis and multi-target regression for effective facility maintenance
 - Developed a structured prediction framework to address the gas turbine monitoring and reliability prediction problem
- Mentor: Dr. Dmitriy Fradkin

Research Intern

May 2014 - Aug 2014

Bosch Research and Technology Center North America, Palo Alto, CA, USA

- Topic: Cost-sensitive decision-theoretic classification/rule mining for automated identification of scraps from manufacturing processes
 - Implemented a software package that supports decision making in class-imbalance and cost-sensitive scenario
- Mentor: Dr. Rumi Ghosh

REFEREED PUBLICATIONS

- H. Jeong, **C. Hong**. READ-ADMET: Retrieval-Enhanced Agent for Drug ADMET Prediction. American Medical Informatics Association (AMIA) Annual Symposium. 2026.
- S. Park, D. Chung, **C. Hong**. Multimodal Forecasting for Commodity Prices Using Spectrogram-Based and Time Series Representations. Association for the Advancement of Artificial Intelligence 2026 Summer Symposium (AAAI-SS). 2026.
- Y. Kim, S. Park, C. Kwon, **C. Hong**. Multimodal Drug Recommendation with Quantum Chemical Molecular Representations. International Conference on Database Systems for Advanced Applications (DASFAA). 2026.
- E. Jung, E. Jeong, U. Lee, Y. Shim, **C. Hong**. Content-Grounded Semantic Identifier Construction for Surveillance Video Indexing. The 12th International Workshop on Big Data Management and Service (BDMS) at DASFAA. 2026.
- C. Hong**, D. Chung, J. Kim, H. Jung. “*Quantum Deepflow: A Quantum-Integrated Forecasting Platform for Strategic Decisions in Raw Material Procurement.*” ACM International Conference on Information and Knowledge Management (CIKM). 2025.
- H. Kim, Y. Wang, M. Ahn, H. Choi, Y. Zhou, **C. Hong**. “*Harnessing EHRs for Diffusion-based Anomaly Detection on Chest X-rays.*” International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI). 2025.
- J. Moon and **C. Hong**. “*Retrieval-Augmented VLMs for Multimodal Melanoma Diagnosis.*” International Skin Imaging Collaboration (ISIC) Skin Image Analysis Workshop at MICCAI. 2025.
- C. Kwon, Y. Kim, S. Park, Y. Lee, **C. Hong**. “*MMM: Quantum-Chemical Molecular Representation Learning for Combinatorial Drug Recommendation.*” International Workshop on Predictive Intelligence In Medicine at MICCAI (PRIME-MICCAI). 2025.
- S. Park, H. Jeong, S.S. Kim, J.Y. Cheong, **C. Hong**. “*Transformer-Based Risk Estimation of HCC in Patients with Chronic Hepatitis B.*” International Workshop on Predictive Intelligence In Medicine at MICCAI (PRIME-MICCAI). 2025.
- J. Moon and **C. Hong**. “*Multimodal Clinical Decision Support for Melanoma Diagnosis Using Retrieval-Augmented Generation and Vision-Language Models.*” IEEE International Conference on Advanced Visual and Signal-Based Systems (AVSS). 2025.
- D. Park, S. Park, C. Lee, H. Choi, **C. Hong**. “*Fine-Grained Video Indexing and Retrieval with Vision-Language Models.*” IEEE International Conference on Advanced Visual and Signal-Based Systems (AVSS). 2025.
- H. Chun, C. Lee, H. Choi, **C. Hong**. “*Bridging the Linguistic Divide: Developing a North-South Korean Parallel Corpus for Machine Translation.*” Pacific Asia Conference on Language, Information, and Computation (PACLIC 38). 2024.
- H. Kim, C.H. Lee, **C. Hong**. “*Transformer for Point Anomaly Detection.*” ACM International Conference on Information and Knowledge Management (CIKM). 2024.
- J. Jung, S. Park, H. Kim, C. Lee, **C. Hong**. “*Artificial Intelligence-Driven Video Indexing for Rapid Surveillance Footage Summarization and Review.*” International Joint Conference on Artificial Intelligence (IJCAI) Demo Track. 2024.

- H. Kim, M. Kim, T. Kim, K. Cho, **C. Hong**. “*AREST: Attention-Based Red-Light Violation Detection for Safety Technology*.” IEEE International Conference on Advanced Video and Signal-Based Surveillance (AVSS). 2024.
- H. Kim, C.H. Lee, **C. Hong**. “*VATMAN: Video Anomaly Transformer for Monitoring Accidents and Nefariousness*.” IEEE International Conference on Advanced Video and Signal-Based Surveillance (AVSS). 2024.
- G. Choi, H.J. Cho, S.S. Kim, J.E. Han, J.Y. Cheong, **C. Hong**. “*Drug Induced Liver Injury Prediction with Injective Molecular Transformer*.” IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI). 2023.
- H. Kim, C.H. Lee, **C. Hong**. “*Crime Scene Detection in Surveillance Videos Using Variational AutoEncoder-Based Support Vector Data Description*.” Asian Conference on Intelligent Information and Database Systems (ACIIDS). 2023.
- G. Choi, H. Yook, J.W. Han, **C. Hong**. “*AWS: GNNs that Aggregate with Self-node Representation for Dehydrogenation Enthalpy Prediction*.” Asian Conference on Intelligent Information and Database Systems (ACIIDS). 2023.
- D. Shin, G. Choi, **C. Hong**, and J.W. Han. “*Surface Segregation Machine-learned with Inexpensive Numerical Fingerprint for the Design of Alloy Catalysts*.” Molecular Catalysis. 2023.
- K. Lee, **C. Hong**, E.H. Lee, W. Yang. “*Comparison of artificial intelligence methods for prediction of mechanical properties*.” IOP Conference Series: Materials Science and Engineering. 2020.
- G. Trivedi, **C. Hong**, E. Dadashzadeh, R. Handzel, H. Hochheiser, S. Visweswaran. “*Identifying incidental findings from radiology reports of trauma patients: An evaluation of automated feature representation methods*.” International Journal of Medical Informatics. 2019.
- C. Hong** and M. Hauskrecht. “*Multivariate Conditional Outlier Detection: Identifying Unusual Input-Output Associations in Data*.” International FLAIRS Conference, Melbourne, FL, USA, 2018.
- C. Hong**, S. Liu, and M. Hauskrecht. “*Detection of Unusual Input-Output Associations*.” SIAM Workshop on Dimension Reduction, Pittsburgh, PA, USA, 2017.
- M. Hauskrecht, I. Batal, **C. Hong**, Q. Nguyen, G. F. Cooper, S. Visweswaran, and G. Clermont. “*Outlier-based detection of unusual patient-management actions: An ICU study*.” Journal of Biomedical Informatics 64: 211-221, 2016.
- C. Hong**, R. Ghosh, and S. Srinivasan. “*Dealing with Class Imbalance Using Thresholding*.” ACM SIGKDD Workshop on Outlier Definition, Detection, and Description on Demand (ODD 4.0), San Francisco, CA, USA, 2016.
- C. Hong** and M. Hauskrecht. “*Multivariate Conditional Outlier Detection and Its Clinical Application*.” AAAI Conference on Artificial Intelligence (Abstract), Phoenix, AZ, USA, 2016.
- C. Hong**, I. Batal, and M. Hauskrecht. “*A Generalized Mixture Framework for Multi-label Classification*.” SIAM International Conference on Data Mining (SDM), Vancouver, BC, Canada, 2015.
- C. Hong** and M. Hauskrecht. “*Multivariate Conditional Anomaly Detection and Its Clinical Application*.” AAAI Conference on Artificial Intelligence (Abstract), Austin, TX, USA, 2015.
- C. Hong**, I. Batal, and M. Hauskrecht. “*A Mixtures-of-Trees Framework for Multi-Label Classification*.” ACM Conference of Information and Knowledge Management (CIKM), Shanghai, China, 2014.
- M. Pakdaman, I. Batal, Z. Liu, **C. Hong**, and M. Hauskrecht. “*An Optimization-based Framework to Learn Conditional Random Fields for Multi-label Classification*.” SIAM International Conference on Data Mining (SDM), Philadelphia, PA, USA, 2014.
- I. Batal, **C. Hong**, and M. Hauskrecht. “*An Efficient Probabilistic Framework for Multi-Dimensional Classification*.” ACM Conference of Information and Knowledge Management (CIKM), Burlingame, CA, USA, 2013.
- C. Hong** and M. Hauskrecht. “*Analysis of Temporal Clinical Patterns Using Hidden Markov Models*.” International Conference on Machine Learning (ICML) Workshop on Clinical Data Analysis, Edinburgh, Scotland, UK, 2012.

PATENT / INVENTION DISCLOSURE

“*Value Addition Dependent Data Mining Techniques for Assembly Lines*.” US Patent No. 10817800.

“*Personal Color Diagnosis Method and System Based on Machine Learning and Augmented Reality*” (머신러닝 및 증강 현실 기반의 퍼스널 컬러 진단 방법 및 시스템). Korean Patent No. 10-2425873.

“*Real-Data-Based Method for Predicting Electric Vehicle Driving Range Using Telematics and BMS Data*” (전기자동차의 텔레매틱스 데이터 및 BMS 데이터를 활용한 실데이터 기반 전기자동차 주행 가능거리 예측 방법). Korean Patent No. 10-2789791.

- “*Device and Method for Predicting Side Effects of Hepatitis B Antiviral Drugs*” (B형 간염 항바이러스제 부작용 예측 장치 및 그 방법). Korean Patent No. 10-2883323.
- “*Demand Forecasting Method Using Optimization Techniques*” (최적화 기법을 이용한 수요 예측 방법). Korean Patent No. 10-2884044.
- “*Pre-Training and Transfer Learning Used Demand Forecasting Apparatus and Its Method*” (사전학습과 전이학습을 이용한 수요 예측 장치 및 그 방법). Application No. 10-2023-0121050. (pending)
- “*Prediction Method Using Pre-Training and Transfer Learning and Target Prediction Framework for Performing the Method*” (사전학습 및 전이학습을 이용한 타겟 예측 방법 및 이를 수행하는 타겟 예측 프레임워크). Application No. 10-2023-0172099. (pending)
- “*Dashcam Video Analysis Network for Enhanced Safe Driving*” (안전운전 강화를 위한 블랙박스 영상 분석 네트워크). Application No. 10-2023-0185288. (pending)
- “*Spectrogram-Enhanced Multimodal Time Series Prediction Method*” (스펙트로그램 강화 기반 멀티모달 시계열 예측 방법). Application No. 10-2026-0005458. (pending)
- “*Spectrogram-Enhanced Multimodal Time Series Prediction System*” (스펙트로그램 강화 기반 멀티모달 시계열 예측 시스템). Application No. 10-2026-0005439. (pending)
- “*AI-based method and system for amending rejected claims in patent prosecution*” (인공지능 기반의 특허 출원절차에 서의 하나 이상의 거절 청구항 보정 방법 및 시스템). Application No. 10-2026-0059930. (pending)
- “*Apparatus and Method for Automatically Classifying Form and Color Based on Stool Images*” (대변 이미지 기반 형태 및 색상 자동 분류 장치 및 방법). Application No. 10-2026-0118243. (pending)
- C. Hong**, D. Fradkin, and A. Chakraborty. “*Multi-Target Regression Using Canonical Correlation Analysis.*” Invention disclosure No. 2015E16593US, Siemens.
- H. Hochheiser, S. Visweswaran, G. Trivedi, **C. Hong**, R. Handzel, and E. Dadashzadeh. “*Automation of Useful Secondary Findings From Radiology and Pathology Reports.*” Invention disclosure No. 04737, University of Pittsburgh.
- M. Hauskrecht, G. Clermont, G. F. Cooper, S. Malakouti, **C. Hong**, Z. Luo, M. P. Barren, and S. Liu. “*Real-time Clinical Monitoring and Alerting System.*” Invention disclosure No. 04995, University of Pittsburgh.

HONORS / AWARDS

Best Paper Award, <i>Korea Computer Congress (KCC)</i>	2026
Best Poster Award, <i>The Liver Week (TLW) 2026</i>	2026
Distinguished Oral Presentation Award, <i>Spring International Convention of the Pharmaceutical Society of Korea</i>	2026
Best Paper Award, <i>Korean Society of Medical and Biological Engineering</i>	2025
Best Presentation Paper Award, <i>Conference on Human and Cognitive Language Technology (HCLT)</i>	2025
Best Paper Award, <i>Annual Symposium of Korea Information Processing Society</i>	2025
Best Paper Award, <i>Korean Society of Medical & Biological Engineering</i>	2024
Best Paper Award, <i>IEEE International Conference on Advanced Video and Signal-Based Surveillance (AVSS)</i>	2024
Best Paper Award, <i>Korea Multimedia Society</i>	2024
Best Presentation Award, <i>The Liver Week (TLW) 2024</i>	2024
Best Student Paper Award, <i>Korea Software Congress (KSC)</i>	2023-2024
Best Paper Award, <i>Korea Computer Congress (KCC)</i>	2023
Best Paper Award, <i>Korean Society of Medical Informatics</i>	2022
Best Capstone Project in AI Award, <i>Korea Multimedia Society</i>	2022
Best Student Paper Award, <i>Korea Computer Congress (KCC)</i>	2020-2023
Best Paper Award, <i>Spring Conference of Korea Entertainment Industry Association</i>	2020

Certification of Achievement, <i>Korean Institute of Information Scientists and Engineers</i>	2019
3rd Prize (DermaQ), <i>IBM BlueHack, IBM and AmerisourceBergen</i>	2018
Travel Grant, <i>Graduate and Professional Student Government, University of Pittsburgh</i>	2016
1st Prize, Research Competition (Area: Artificial Intelligence), <i>University of Pittsburgh</i>	2015
Doctoral Consortium Scholarship, <i>Association for the Advancement of Artificial Intelligence</i>	2015
Travel Grant, <i>Dietrich School of Arts and Science, University of Pittsburgh</i>	2015 - 2016
Travel Grant, <i>Society for Industrial and Applied Mathematics</i>	2015
Travel Grant, <i>Department of Computer Science, University of Pittsburgh</i>	2012 - 2016
Travel Grant, <i>National Science Foundation</i>	2012, 2015

PRESENTATIONS

Intro to AI Capstone Design Project for High School Students <i>Gyeongsangbuk-do Office of Education AI Education Center, Uiseong, Gyeongbuk, Korea</i>	Jul 2026
Advancing Multimodal Medical AI through Global Research Collaboration <i>The 6th Handong BK21 AI Workshop, Jeju, Korea</i>	Jun 2026
A Tour of Deep-Learning NLP and Negation <i>Department of English Language and Literature, Kyung Hee University, Seoul, Korea</i>	Jun 2026
Designing and Implementing Introductory Courses for Responsible Engineering and AI Education <i>Spring Conference of Korea Association for Christian Scholarship, Cheonan, Chungnam, Korea</i>	May 2026
Recent Advances in Medical AI and Its Applications in Clinical Practice <i>Daegu Gyeongbuk Hospital Association, Daegu, Korea</i>	May 2026
Conducting Research in an Undergraduate-Focused University <i>New Faculty Fellowship, Handong Global University, Pohang, Gyeongbuk, Korea</i>	May 2026
Machine Learning as an Integrative Layer for Biomedical Intelligence <i>National Institute of Health, Osong, Chungbuk, Korea</i>	Apr 2026
Designing AI Systems for Human Decision-Making in Safety-Critical Domains <i>JJANT-IT, Jeonbuk National University, Jeonju, Jeongbuk, Korea</i>	Mar 2026
Research at HAIL: Designing AI for the Holistic Restoration of Human Value <i>Human-AI Workshop, National Chengchi University, Taipei, Taiwan</i>	Mar 2026
An Overview of Research at HAIL <i>Impactive AI, Seoul, Korea</i>	Jul 2025
A Tutorial on Outlier Detection <i>The 4th Co-week Academy, Pyeongchang, Gangwon, Korea</i>	Jul 2025
AI for the Holistic Restoration of Human Value: From Healthcare to Mobility and Meaning <i>The 5th Handong BK21 AI Workshop, Pohang, Gyeongbuk, Korea</i>	Jun 2025
Video Anomaly Detection in DeepView <i>GMDSOFT, Seongnam, Gyeonggi, Korea</i>	May 2024
Generation of Parallel Corpus for South and North Korean Languages <i>Korea Institute of Science and Technology Information (KISTI), Daejeon, Korea</i>	Dec 2023
Development of an AI-Driven Analytics System for North Korean Literature <i>Korea Institute for National Unification, Seoul, Korea</i>	Nov 2023
Development of KUBiC (Korean Unification Bigdata Center) <i>Handong Institute for Peace and Reconciliation, Pohang, Gyeongbuk, Korea</i>	Nov 2023

Machine Learning as a Strategic Tool to Support Business Operations <i>Startup Training for African Revival (STAR), St. Paul's University, Limuru, Kenya</i>	Feb 2023
Research on Machine Learning Model Development, Share, and Code Quality Measurement Methods <i>Korea Institute of Science and Technology Information (KISTI), Daejeon, Korea</i>	Nov 2022
Case Studies on Computer Science Education at Handong Using Korean Unification as a Subject <i>Korea Computer Congress (KCC), Korean Institute of Information Scientists and Engineers, Jeju, Korea</i>	Jun 2022
Online Machine Learning Environment for Enhancing Code Quality and Data Usability <i>Korea Institute of Science and Technology Information (KISTI), Daejeon, Korea</i>	Dec 2021
Model Development Environment for Reproducible Machine Learning Artifact Archiving <i>Korea Institute of Science and Technology Information (KISTI), Daejeon, Korea</i>	Dec 2020
An Overview of Outlier Detection Techniques for Complex Data <i>Hancom, Seongnam, Gyeonggi, Korea</i>	Oct 2020
Outlier Detection in Time Series Data <i>Pohang University of Science and Technology (POSTECH), Pohang, Gyeongbuk, Korea</i>	Oct 2020
Building and Application of Medical Decision-Making Models Using Multi-Output Prediction <i>Korea Computer Congress (KCC), Korean Institute of Information Scientists and Engineers, Jeju, Korea</i>	Jun 2019
Artificial Intelligence in Medicine and Clinical Outlier Detection <i>School of Life Science, Handong Global University, Pohang, Gyeongbuk, Korea</i>	Apr 2019
Machine Learning with Multiple Structured Targets <i>POSCO, Pohang, Gyeongbuk, Korea</i>	Mar 2019
Multivariate Data Modeling and Its Applications to Conditional Outlier Detection <i>Department of Computer Science, University of Pittsburgh, Pittsburgh, PA, USA</i>	Aug 2017
Detection of Unusual Input-Output Associations (Poster) <i>SIAM Workshop on Dimension Reduction, Pittsburgh, PA, USA</i>	Jul 2017
Improving Multi-label Classification Accuracy Using a Generalized Mixture Framework (Poster) <i>Advancing Research through Computing (ARC) 2017, Pittsburgh, PA, USA</i>	Mar 2017
Detection of Unusual Input-Output Patterns in Electronic Health Records (Poster) <i>NSF Data Science Workshop, Seattle, WA, USA</i>	Aug 2015
Multi-Target Prediction <i>Siemens Corporate Technology, Princeton, NJ, USA</i>	Jul 2015
Multivariate Conditional Outlier Detection (Poster) <i>SIAM International Conference on Data Mining (SDM) Doctoral Forum, Vancouver, BC, Canada</i>	Apr 2015
Multivariate Conditional Outlier Detection in Clinical Domain <i>Annual Computer Science Day 2015, University of Pittsburgh, Pittsburgh, PA, USA</i>	Mar 2015
Multivariate Conditional Anomaly Detection and Its Clinical Application <i>Association for the Advancement of Artificial Intelligence (AAAI) Doctoral Consortium 2015, Austin, TX, USA</i>	Jan 2015
Multi-label Classification <i>Department of Computer Science, University of Pittsburgh, Pittsburgh, PA, USA</i>	Nov 2014
Building Decision Theoretic Classifiers for Imbalanced Datasets <i>Bosch Research and Technology Center North America, Palo Alto, CA, USA</i>	Aug 2014
Multi-Dimensional Classification with Conditional Tree-structured Bayesian Networks <i>Dietrich School of Arts and Sciences Grad Expo 2014, Pittsburgh, PA, USA</i>	Mar 2014
Multi-Label Classification with Conditional Tree-structured Bayesian Networks (Poster) <i>Google Regional PhD Summit 2014, Pittsburgh, PA, USA</i>	Jan 2014
Classification of Temporal Clinical Patterns Using Hidden Markov Models (Poster) <i>Google Regional PhD Summit 2013, Pittsburgh, PA, USA</i>	Mar 2013

TEACHING EXPERIENCES

Instructor, *School of Computer Science and Electrical Engineering, Handong Global University, Pohang, Korea*

- ITP40010 Machine Learning
- ECE20010/ITP20001 Data Structures
- ECE10020 Intro to Engineering Design
- ECE10002 C Programming
- ECE30030/ITP30010 Database System
- ITP20004 Open-source Software Labs
- ECE10003 C Programming Labs

Instructor, *Department of Chemical Engineering, Pohang University of Science and Technology, Pohang, Korea*

- CHEB801Q Machine Learning Practicum for Engineering Students

Instructor, *Data Science Camp in Cambodia, UNESCO-UNITWIN, Phnom Penh, Cambodia*

- Machine Learning

Teaching Assistant, *Department of Computer Science, University of Pittsburgh, Pittsburgh, PA, USA*

- CS2750 Machine Learning
- CS1571 Introduction to Artificial Intelligence
- CS1501 Algorithm Implementation
- CS2710 Foundations of Artificial Intelligence
- CS1502 Formal Methods in Computer Science
- CS401 Intermediate Programming Using Java

VOLUNTEERING / EXTRACURRICULAR ACTIVITY

Reviewer , <i>IEEE Journal of Biomedical and Health Informatics</i>	2026
Reviewer , <i>ACM Transactions on Information Systems</i>	2026
Reviewer , <i>IEEE International Conference on Data Mining (ICDM)</i>	2026
Reviewer , <i>International Conference on Medical Image Computing and Computer Assisted Intervention</i>	2026
Reviewer , <i>ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)</i>	2026-2027
Reviewer , <i>Learning on Graphs Conference (LoG)</i>	2025-2026
Reviewer , <i>IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI)</i>	2025-2026
Reviewer , <i>ACM International Conference on Information and Knowledge Management (CIKM)</i>	2025-2026
Reviewer , <i>Public Library of Science (PLOS)</i>	2024-2026
Reviewer , <i>IEEE International Symposium on Biomedical Imaging (ISBI)</i>	2024-2025
Session Chair , <i>International Joint Conference on Artificial Intelligence (IJCAI)</i>	2024
Reviewer , <i>IEEE Access</i>	2023-2024
Reviewer , <i>ISPRS International Journal of Geo-Information</i>	2023
Reviewer , <i>Korean Institute of Information Scientists and Engineers (KIISE)</i>	2020-2026
Reviewer , <i>Multidisciplinary Digital Publishing Institute (MDPI)</i>	2020-2024
Reviewer , <i>International Joint Conference on Artificial Intelligence (IJCAI)</i>	2018
Reviewer , <i>AAAI Conference on Artificial Intelligence (AAAI)</i>	2016-2018
Reviewer , <i>IEEE Transactions on Knowledge and Data Engineering (TKDE)</i>	2014-2016
Reviewer , <i>IEEE Transactions on Big Data (TBD)</i>	2016
Reviewer , <i>Journal of Biomedical Informatics (JBI)</i>	2015, 2024
Reviewer , <i>Pattern Recognition Letters (PRLETTERS)</i>	2015
Reviewer , <i>Conference on Artificial Intelligence in Medicine (AIME)</i>	2015
Student Volunteer , <i>International Conference on Machine Learning (ICML), Edinburgh, Scotland, UK</i>	2012
Workshop Director , <i>Engineering Design with the Other 90% (D90), Wonju, Korea</i>	2010